

Emily Herbert

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Education

- 2025 **Art and French Language Intensive Training**, *Atelier de Sevres*, Paris, France.
- 2021 **MS in Computer Science**, *The University of Massachusetts Amherst*, Amherst, MA.
- 2018 **BS in Computer Science**, *Trinity University*, San Antonio, TX.

Skills

Languages, Proficient: Rust, Sway, Scala, JavaScript, TypeScript, Python

Familiarity: Solidity, Haskell, OCaml, Idris, C, C#, C++, R, Java, Greenfoot, ScalaFX, JavaFX.

Tools, Cargo, Yarn, Kubernetes, Docker, OpenWhisk, Google Cloud Platform, Unity, Helm, Sbt .

Specialties, Compilers, Programming Languages, Blockchain, Smart Contracts, Software Design, Systems Design, Algorithms, Cloud Computing.

Relevant Experience

Aug 2021 - **Fuel Labs**.

May 2023 Compiler Engineer - Rust

Designed and implemented a compiler for the Sway smart contract programming language, focusing on friendly interactivity with developers. Built tools to integrate with the FuelVM and the Fuel ecosystem.

Talks & Panels.

Blockchain Language Design panel. *Layer 2 Day at EthDenver*. 2023. [[recording](#)]

Introduction to Fuel: Let's Get Modular talk. *Celestia's Modular Fellows*. 2022. [[slides](#)]

Beyond Monolithic with Fuel: the Fastest Modular Execution Layer talk. *HackMoney*. 2022. [[slides](#)]

Scaling Execution: Optimistic panel. *The Modular Summit*. 2022.

Developing Smart Contracts in Sway talk. *Layer2Amsterdam*. 2022. [[slides](#)]

Sway: A Rust-based Smart Contract Language talk. *EthDenver*. 2022. [[slides](#)]

The Rollup Developer Experience panel. *EthDenver*. 2022. [[recording](#)]

May 2021 - **Google, Madison, WI**.

Aug 2021 Software Engineering Intern

Implemented load balancing in a library meant to interface with the network card and perform RPC-like operations using RMA, achieved by integrating two existing early-development libraries together. Contributed to app design on a Google 2023 project.

June 2018 - **Northeastern University, University of Massachusetts Amherst, Boston, MA, Amherst, MA**.

May 2021 Researching programming language and systems tools for serverless computing. prl.ccs.neu.edu. plasma-umass.org.

Talks & Publications.

Emily Herbert and Arjun Guha. A Language-based Serverless Function Accelerator. 2021. [[preprint](#), [repo](#)]

A Language-based Serverless Function Accelerator talk. *Google PhD Intern Research Conference*. 2021.

A Language-based Serverless Function Accelerator talk. *Cornell CAPRA Lab*. 2020. [[slides](#)]

June 2018 - **University of Massachusetts Amherst, Amherst, MA**.

May 2019 Researching deep learning methods for simulation input modeling. dbgroupp.cs.umass.edu

Talks & Publications.

Wang Cen, Emily A. Herbert, and Peter J. Haas. NIM: Modeling and Generation of Simulation Inputs via Generative Neural Networks. *Winter Simulation Conference*. 2020. [[paper](#)] **Best Contributed Theoretical Paper Finalist**

Wang Cen, Emily A Herbert, Peter J Haas. Generative Neural Networks for Simulation Input Modeling. *SCS Summer Simulation Conference*. 2019. [[extended abstract](#)]

Emily A Herbert, Wang Cen, and Peter J Haas. NIM: Generative Neural Networks for Simulation Input Modeling. *SCS Summer Simulation Conference*. 2019. [[short paper](#), [slides](#)]

June 2017 - **National Aeronautics and Space Administration (NASA), Langley, VA**.

Aug 2017 NASA Internships, Fellowships, and Scholarships (NIFS) Intern

Contributed to the NASA Safeguard autonomous drone geofencing project. Designed and implemented system for on-board flight control of GPS devices. Refactored code from previous NASA flight missions to meet current mission standards.